

R&D INTENSITY, DIGITAL TRANSFORMATION, AND EXPORT COMPETITIVENESS:

EVIDENCE FROM INDIAN FIRMS

An Empirical Analysis Using CMIE Prowess Firm-Level Data (2014–2025)

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ABSTRACT

This study examines the relationship between research and development (R&D) intensity, digital infrastructure investment, and export performance among Indian firms using firm-level data drawn from the CMIE Prowess database for the period 2014–2025. Two research questions are addressed. The first concerns whether R&D intensity, technology licensing (royalty payments), and sectoral classification are associated with export volume, export intensity, export participation, and the probability of service exports. The second concerns whether a composite Digital Intensity Index, constructed from five categories of digital expenditure, is associated with export intensity and export participation, and whether this relationship differs between manufacturing and services firms. The empirical models combine ordinary least squares (OLS) and logistic regression, with industry fixed effects and a pre-/post-pandemic period split (2014–2019 and 2020–2025) employed to capture structural heterogeneity. The results indicate that sectoral classification, the intensity of imported raw materials, and salary expenditure are the most consistent and statistically reliable predictors of export performance, whereas R&D intensity and the Digital Intensity Index exhibit directionally positive but generally statistically weak associations with export outcomes. The findings suggest that the influence of innovation and digital investment on export competitiveness is contingent upon industry context, global supply chain integration, and workforce composition, rather than operating as uniform, stand-alone drivers of export success. The study concludes with implications for corporate strategy, innovation policy, and the methodological design of future firm-level export research.

Keywords: Export competitiveness; R&D intensity; Digital Intensity Index; Indian firms; CMIE Prowess; logistic regression; firm-level analysis

SECTION 1: INTRODUCTION TO THE STUDY

Export competitiveness is widely regarded as a function of firm-level capabilities, including innovation capacity, technological investment, and integration into global value chains. A long-standing question in international trade research is whether firms become more productive because they export, or whether already-productive firms self-select into export markets. Resolving this question matters directly for policy, because if technological capability is what allows firms to compete internationally, then innovation and digital investment become natural levers for export promotion; if, instead, exporting itself is the driver of subsequent technology adoption, then trade-facilitation and market-access policies take precedence over supply-side innovation subsidies.

This study investigates these relationships among Indian firms through two related research questions. The first concerns the relationship between R&D intensity, technology licensing, and export performance, with particular attention to whether this relationship is amplified in technology-intensive sectors. The second concerns the relationship between digital infrastructure investment — captured through a composite Digital Intensity Index — and export performance, with particular attention to whether this relationship differs between manufacturing and services firms.

The analysis draws on firm-year observations from the CMIE Prowess database, separated into two periods, 2014–2019 and 2020–2025, allowing an examination of firm behaviour before and after the COVID-19 pandemic, a period associated with accelerated digital adoption among Indian businesses. India provides a particularly relevant setting for this inquiry. The country has built export strength in both technology-intensive manufacturing (pharmaceuticals, chemicals, electronics) and knowledge-intensive services (information technology and business process outsourcing), while simultaneously pursuing an active industrial and digital policy agenda intended to raise the technological content of its export base. Understanding whether, and under what conditions, R&D and digital investment translate into measurable export gains is therefore of direct relevance to Indian policymakers, corporate strategists, and the broader academic literature on technology and trade.

The remainder of this manuscript is organised as follows. Section 2 reviews the relevant literature on technological capability, R&D, digital transformation, and export performance, with particular attention to the Indian context. Section 3 identifies the research gap and states the research objectives addressed by this study. Section 4 describes the data, variable construction, and statistical methodology. Section 5 reports and discusses the empirical findings, including the economic significance of statistically reliable results. Section 6 discusses the implications of the research for corporate strategy, policy, and methodology. Section 7 concludes, noting the limitations of the study and avenues for future research.

SECTION 2: LITERATURE REVIEW

2.1 Technology, Innovation, and Export Performance: Theoretical Foundations

The literature on technology and exports begins with a foundational question: do firms export because they are productive, or do they become more productive after they begin exporting? Wagner (2007) reviewed firm-level evidence from across the world and found that high-productivity firms are systematically more likely to export, establishing self-selection as a robust empirical regularity and shifting the research agenda toward explaining how firms acquire the comparative advantage that allows them to self-select. Melitz (2003) provided a complementary theoretical foundation, showing that trade liberalisation reallocates resources toward the most productive firms within an industry, which are precisely the firms able to absorb the fixed costs of entering export markets. Bernard and Jensen (1999) and Clerides, Lach, and Tybout (1998) extended this debate empirically, finding mixed evidence for “learning-by-exporting” — the proposition that the act of exporting itself, through exposure to foreign buyers and competitors, induces subsequent productivity and technology gains.

Beyond the direction of causality, a second strand of the literature asks which dimensions of technology matter most for export performance, and whether this varies systematically with industry characteristics. Montobbio and Rampa (2005), studying nine large developing economies, found that export performance in high-technology industries is explained primarily by R&D spending and foreign direct investment, whereas cost factors dominate in low-technology industries — implying that the technology–export relationship is industry-contingent rather than universal. Flor and Oltra (2005), examining Spanish manufacturing firms, found that practical innovation activities such as engineering and design contributed more to export performance than R&D expenditure itself, suggesting that how technology is deployed matters as much as how much is spent on it. Taken together, this foundational literature establishes that technology is associated with export success, but that the strength and form of this association is neither uniform across industries nor straightforwardly causal — a tension that motivates much of the subsequent Indian-context literature reviewed below.

2.2 R&D Intensity and Export Behaviour in the Indian Context

A substantial body of Indian firm-level research has examined whether R&D investment and imported technology predict export behaviour. Kumar and Siddharthan (1994) and

the subsequent extension by Kumar and Srinivasan (2007) found that both indigenous and imported technological capabilities were strong predictors of exporting, including in industries not conventionally regarded as technology-intensive, with mid-sized enterprises emerging as particularly active exporters — evidence of meaningful heterogeneity in the technology–export relationship by firm size. Bhat and Narayanan (2009), focusing on India's basic chemical industry, similarly found that firm size and technological effort, both home-grown and imported, were the strongest predictors of export behaviour once more complex empirical specifications were used. Hasan and Raturi (2003) reached a complementary conclusion using a broader sample of Indian manufacturing firms, reporting a positive association between technology investment and export performance. Pradhan (2010) extended this line of work temporally, showing that Indian manufacturing firms with stronger R&D capabilities were not only more likely to enter export markets but also more likely to sustain export growth over time.

Not all evidence points in the same direction, however. Basant and Fikkert (1996) found that R&D investment functioned for many Indian manufacturers as a means of sustaining an existing export presence rather than as a driver of new export growth — a “safety net” interpretation in which technology investment protects market position rather than expanding it. More directly, Basant, Mishra, and Saha (2016), studying Indian manufacturers without foreign affiliates, found no statistically significant difference in technology expenditure between exporters and non-exporters, a result that complicates any simple narrative in which R&D investment is a universal precondition for exporting. This mixed evidence underscores the importance of rigorous, well-specified statistical analysis — including appropriate controls for firm heterogeneity — before inferring a causal or even a robust associative link between R&D spending and export status.

A further consideration in this literature concerns timing. Aw, Roberts, and Winston (2007) showed that firms increase investment in R&D and worker training in the period immediately preceding export entry, implying that technology investment among Indian “export starters” may reflect deliberate preparation for international competition rather than a contemporaneous response to exporting itself — a pattern that has direct implications for how cross-sectional, single-period regressions of the kind employed in much of this literature should be interpreted. Examining causality from the opposite direction, Dzhumashev, Mishra, and Smyth (2016), studying the Indian IT sector, found that exporting firms invested more in R&D and innovation in response to competitive pressure from global markets, and that this investment in turn improved firms' survival prospects internationally. Tyagi (2022) found a parallel feedback effect in Indian manufacturing, with export intensity acting as a statistically significant determinant of subsequent R&D investment in R&D-intensive industries. Sinha, Mishra, and Patel (2019), using a quarter-century panel of Indian firms, confirmed that the income benefits of R&D expenditure were concentrated among outward-oriented, high-performing firms,

while Reddy (2024) found that innovation investment and exporting were mutually reinforcing even among Indian SMEs. The most recent synthesis, by Sahoo, Vadakoot Sathees, and Anand (2025), found that R&D investment, human capital development, and imported technology jointly predicted export success in Indian manufacturing more strongly than any single dimension alone, with the benefit of imported technology depending on a firm's internal absorptive capacity.

2.3 Digital Transformation and Export Competitiveness

A more recent strand of the literature shifts attention from R&D to digital infrastructure as a determinant of export competitiveness. The OECD (2019) and the Economic Research Institute for ASEAN and East Asia (ERIA, 2020) both document a broader Asia-Pacific pattern in which digitalisation — the adoption of software, e-commerce, and internet-enabled business processes — lowers the fixed and variable costs of cross-border trade, with ERIA's analysis associating each one-percentage-point increase in digital intensity with an average increase in exports of approximately 0.16 percent. Sandu and Ciocanel (2014), working at the country level across European Union member states, found that private R&D spending and knowledge-intensive human capital were stronger drivers of high-technology export competitiveness than exchange rates, labour costs, or market size, lending support to a view of digital and knowledge capital as complements to, rather than substitutes for, traditional R&D.

Financing structure has also been shown to mediate the technology–export relationship in the Indian services context specifically. Yang, Milner, Lancheros, and Gunessee (2020) found that internal financing broadly supported investment in both technology and exporting among Indian services firms, whereas access to equity markets had a more specific effect on modern, knowledge-intensive service exporters — evidence that the digital and technological investment decisions of services exporters cannot be fully understood without reference to how those investments are financed. Despite this growing body of evidence, the literature on digital infrastructure and Indian export performance remains considerably thinner than the corresponding R&D literature, and — as discussed in Section 3.1 — has not yet produced a composite, multi-category measure of digital investment comparable to the R&D intensity measures used throughout Section 2.2.

2.4 Methodological Approaches in Firm-Level Export Research

The methodological toolkit used to study the technology–export relationship has evolved considerably over the period covered by this literature. Early studies relied primarily on cross-sectional ordinary least squares regression of export outcomes on technology measures. Because exporting firms are not a random sample — they tend to be larger, more productive, and more capital-intensive than non-exporters — later studies increasingly adopted methods designed to address this selection problem, including propensity score matching and fixed-effects panel estimation, which compare similar firms before and after export entry rather than relying on simple cross-sectional comparisons. Logistic regression has become the standard approach for binary export-participation outcomes, given the non-continuous nature of the dependent variable, while OLS remains appropriate for continuous outcomes such as export intensity or log export value.

Notwithstanding these methodological advances, important limitations persist across the literature reviewed in Sections 2.1 through 2.3. Few studies employ instrumental variables or natural experiments capable of isolating causal direction; fewer still incorporate firm fixed effects, owing to the practical difficulty of constructing consistent firm identifiers across long panels in databases such as CMIE Prowess; and almost none of the literature on Indian firms constructs a composite digital investment index analogous to the R&D intensity measures that dominate Section 2.2. These gaps directly motivate the data and methodology adopted in Section 4 of this study.

SECTION 3: RESEARCH GAP AND OBJECTIVES

3.1 Research Gap

Read as a whole, the literature reviewed in Section 2 establishes a broad narrative in which technological capability is closely associated with export success among Indian firms, spanning two decades of research from Wagner’s (2007) foundational productivity–exports work to Sahoo, Vadakoot Sathees, and Anand’s (2025) recent synthesis of R&D, human capital, and imported technology. However, several gaps in this literature remain unaddressed.

First, India’s services sector — despite its substantial and growing share of national export earnings, particularly in information technology and business process services — remains underrepresented relative to manufacturing in the empirical literature on technology and exports. Second, digital infrastructure investment as a distinct, multi-dimensional driver of export competitiveness has been considerably less studied than R&D investment, and no prior Indian study has combined software, IT-enabled services, internet service provider expenses, data centre costs, and satellite/VSAT charges into a

single composite Digital Intensity Index. Third, the existing literature offers limited evidence on how the technology–export relationship behaved across the structural break associated with the COVID-19 pandemic, a period widely understood to have accelerated digital adoption among Indian businesses. Fourth, much of the existing evidence — as discussed in Section 2.4 — does not adequately separate sectoral, supply-chain, and workforce explanations of export performance from the effects of R&D and digital investment narrowly defined, raising the possibility that single-variable studies have overstated the independent contribution of technology investment relative to these other firm characteristics.

This study addresses these gaps by jointly estimating the relationship between R&D intensity, royalty/technology licensing, a newly constructed Digital Intensity Index, and multiple export outcomes, using a large firm-level sample that spans the pre- and post-pandemic periods and explicitly distinguishes manufacturing from services firms.

3.2 Research Objectives

In light of the research gap identified above, this study is guided by two research questions and four associated objectives:

Research Question 1: Are R&D intensity and royalty/technology-licensing payments associated with export volume, export intensity, export participation, and the probability of service exports among Indian firms, and is this relationship stronger in technology-intensive sectors?

Research Question 2: Is a composite Digital Intensity Index associated with export intensity and export participation among Indian firms, and does this relationship differ between manufacturing and services firms?

Correspondingly, the specific objectives of the study are:

- To estimate the association between R&D intensity, royalty payments, and firm-level export outcomes (export volume, export intensity, export participation, and service-export probability) using firm-year data from the CMIE Prowess database.
- To test whether the R&D–export relationship is amplified in technology-intensive sectors (electronics, software/IT, and R&D services) through an interaction specification.
- To construct a composite Digital Intensity Index from five categories of digital expenditure and estimate its association with export intensity and export participation.

- To test whether the digital intensity–export relationship differs systematically between manufacturing and services firms, and to compare these relationships across the pre-pandemic (2014–2019) and post-pandemic (2020–2025) periods.

SECTION 4: DATA AND METHODOLOGY

4.1 Sample Selection and Data

The data for this study are drawn from the CMIE Prowess database, which provides detailed financial information for Indian companies based on audited annual reports. Prowess records firm-level items including R&D expenses, royalty payments, software costs, information technology (IT) spending, and export receipts, making it a suitable source for examining the relationship between innovation, digital investment, and export performance.

The study covers two periods, 2014–2019 and 2020–2025, allowing a comparison of firm behaviour before and after the COVID-19 pandemic, a period regarded as a turning point for digital adoption among Indian businesses. Observations with missing values for Total Income, Total Assets, Total Expenses, or National Industrial Classification (NIC) Code were dropped, as these variables are essential to the analysis. After data cleaning, the 2020–2025 sample comprised 8,808 firm-year observations, while the 2014–2019 sample comprised 473 firm-year observations. The substantially smaller size of the earlier sample reflects limited reporting of digital expenditure items during that period: several of the variables required to construct the Digital Intensity Index were sparsely reported in the earlier years of the database, and applying consistent completeness criteria across both periods generated substantial attrition in the 2014–2019 sample.

Table 1. Sample Construction Summary

Stage	2014–2019	2020–2025
Initial observations (pre-cleaning)	19,032	19,027
Final sample (firm-year observations)	473	8,808

Note: counts of observations removed at each intermediate cleaning step (missing financial values versus missing industry classification) were not separately retained in the underlying data-cleaning process and are not estimated here in order to avoid reporting fabricated figures; only the initial pre-cleaning and final cleaned sample sizes are reported.

The dataset spans more than 70 two-digit NIC industries. For Research Question 2, firms are grouped into two broad categories: Manufacturing (NIC < 5000) and Services (NIC ≥ 5000). This classification is used to test whether digital investment is associated differently with exports in services relative to manufacturing. CMIE Prowess primarily covers large and medium-sized listed companies; the implications of this coverage limitation for the generalisability of the findings are discussed in Section 7.1.

4.2 Variable Construction

Two sets of variables are used, corresponding to the two research questions.

For Research Question 1 (R&D and exports), the dependent variables are log-transformed total exports, export intensity (exports divided by total income), and a binary dummy variable indicating whether a firm exported at all, together with a binary indicator of net service-exporter status. The main explanatory variables are R&D intensity (R&D expenses as a share of total income) and royalty intensity (royalty and technical know-how payments as a share of total income). A technology-sector dummy is constructed using NIC codes 26 (electronics), 62 (software/IT), and 72 (R&D services), to examine whether the R&D–export relationship is stronger in technology-intensive industries.

Equations A.1–A.3 — Dependent Variable Construction

$$TE_i = EG_i + ES_i \quad [total\ exports] \quad (A.1)$$

$$\ln TE_i = \ln(1 + EG_i + ES_i) \quad [log-transformed\ export\ volume] \quad (A.2)$$

$$\tau_i = TE_i / TI_i \quad [export\ intensity] \quad (A.3)$$

where:

TE_i = total exports of firm i (INR million)

EG_i = export of goods, free-on-board (FOB) value

ES_i = export of services

$\ln TE_i$ = natural logarithm of total exports (log1p transformation retains zero-export observations)

τ_i = export intensity: total exports as a fraction of total income

Ti_i = total income of firm i (INR million)

Equations A.6–A.9 — Intensity Variable Construction

$$\rho_i = RD_i / (TI_i + 1) \quad [R\&D\ intensity] \quad (A.6)$$

$$\theta_i = ROY_i / (TI_i + 1) \quad [royalty\ intensity] \quad (A.7)$$

$$\sigma_i = SW_i / (TI_i + 1) \quad [software\ intensity] \quad (A.8)$$

$$\mu_i = IRM_i / (TI_i + 1) \quad [imported\ raw\ material\ intensity] \quad (A.9)$$

where:

Ti_i = total income of firm i (INR million)

RD_i = R&D expenses of firm i (set to 0 where missing)

ROY_i = royalties and technical know-how fees of firm i (set to 0 where missing)

SW_i = software charges of firm i (set to 0 where missing)

IRM_i = imported raw materials consumed by firm i (set to 0 where missing)

$+ 1$ = small-constant regularisation preventing division by zero for zero-income observations

For Research Question 2 (digital infrastructure and exports), a Digital Intensity Index (DII) is constructed by standardising five cost items — software charges, IT-enabled services, internet service provider (ISP) expenses, data centre costs, and VSAT/satellite charges — and averaging them into a single composite score. This approach mitigates the multicollinearity that would arise from including all five items as separate explanatory variables in the same model.

Equations A.11a–b — Digital Intensity Index (DII) Construction

$$z_{ik} = (x_{ik} - \mu_k) / \sigma_k, \quad k = 1, 2, 3, 4, 5 \quad [standardisation] \quad (A.11a)$$

$$DII_i = (1/5) \times \sum_{k=1}^5 z_{ik} \quad [composite index] \quad (A.11b)$$

where:

z_{ik} = standardised (z-score) value of digital expenditure component k for firm i

x_{ik} = raw value of component k for firm i (INR million)

μ_k = sample mean of component k across all firms

σ_k = sample standard deviation of component k

$k = 1$ = software charges

$k = 2$ = IT-enabled services charges

$k = 3$ = internet service provider (ISP) expenses

$k = 4$ = data centre, web-hosting, and co-hosting expenses

$k = 5$ = VSAT, satellite links, and communication expenses

DII_i = composite Digital Intensity Index for firm i : arithmetic mean of five standardised components

Standardisation assigns equal weight to each component regardless of raw monetary scale. Aggregation into a single index avoids multicollinearity that would arise from entering all five items separately. The resulting index is mean-centred with cross-sectional standard deviation ≈ 1 .

Control variables common to both research questions include log total assets (a measure of firm size), imported raw material intensity, software charges, salary and wage expenditure, and industry fixed effects defined at the two-digit NIC level.

4.3 Statistical Methodology and Rationale for Test Selection

Seven empirical models are estimated, combining ordinary least squares (OLS) for continuous outcomes and logistic regression for binary outcomes. Industry fixed effects (two-digit NIC) are included throughout, and results are reported separately for the 2014–2019 and 2020–2025 periods to capture structural differences before and after the COVID-19 shock; this period-split approach is distinct from, and should not be interpreted as equivalent to, a full annual year-fixed-effects specification, which could not be consistently estimated across all model specifications given the retained statistical output.

4.3.1 Log Exports Model (OLS)

Equation B.1 — Log Exports Model (OLS), Research Questions 1 & 2

$$\ln TE_i = \beta_0 + \beta_1 \ln RD_i + \beta_2 s_i + \beta_3 \sigma_i + \beta_4 \mu_i + \sum_k \gamma_k \text{Industry}_{ik} + \varepsilon_i \quad (\text{B.1})$$

where:

- $\ln TE_i$ = $\ln(1 + \text{goods exports}_i + \text{services exports}_i)$ [dependent variable]
- $\ln RD_i$ = $\ln(1 + \text{R\&D expenditure}_i)$ [log-transformed R&D; expected sign: positive]
- s_i = $\ln(1 + \text{total assets}_i)$ [firm-size control]
- σ_i = $\text{software charges}_i / (\text{total income}_i + 1)$ [software intensity]
- μ_i = $\text{imported raw materials}_i / (\text{total income}_i + 1)$ [GVC integration]
- γ_k = industry fixed-effect coefficient for two-digit NIC sector k
- $\beta_0 - \beta_4$ = OLS coefficients estimated by least squares
- ε_i = stochastic error term; assumed $E[\varepsilon_i] = 0$

The log–log specification allows β_1 to be interpreted as the approximate export elasticity with respect to R&D expenditure: a 1% increase in R&D spending is associated with a $\beta_1\%$ change in export volume, *ceteris paribus*.

OLS is used because the dependent variable, log exports, is continuous; the log-log specification allows coefficients to be interpreted approximately as elasticities. R&D expenditure, firm size, and imported raw material intensity are each expected to carry positive coefficients, reflecting innovation-driven product quality, the capacity of larger firms to bear the fixed costs of exporting, and global value chain participation, respectively.

4.3.2 Export Intensity Model (OLS)

Equation B.2 — Export Intensity Model (OLS)

$$\tau_i = \beta_0 + \beta_1 \rho_i + \beta_2 \theta_i + \beta_3 s_i + \text{Controls}_i + \sum_k \gamma_k \text{Industry}_{ik} + \varepsilon_i \quad (\text{B.2})$$

where:

- τ_i = $\text{total exports}_i / \text{total income}_i$ [export intensity; dependent variable]
- ρ_i = $\text{R\&D expenses}_i / (\text{total income}_i + 1)$ [R&D intensity; expected sign: positive]
- θ_i = $\text{royalties and technical fees}_i / (\text{total income}_i + 1)$ [royalty intensity; expected sign: positive]
- s_i = $\ln(1 + \text{total assets}_i)$ [firm-size control]
- Controls_i = vector including imported raw material intensity (μ_i) and software intensity (σ_i)
- γ_k = industry fixed-effect for two-digit NIC sector k
- ε_i = error term; assumed i.i.d. with zero mean

OLS is used as export intensity is a continuous, bounded ratio variable. R&D intensity and royalty intensity are expected to carry positive coefficients, reflecting the proposition that innovation capability and technology licensing raise the share of output sold abroad.

4.3.3 Export Participation Model (Logit — Royalty Intensity)

Equation B.3 — Export Participation Model (Logit)

$$Pr(D_i^* = 1) = \Lambda(\beta_0 + \beta_1 \theta_i + \beta_2 s_i + \beta_3 \sigma_i + \beta_4 \mu_i) \quad (B.3)$$

$$\Lambda(u) = e^u / (1 + e^u) = 1 / (1 + e^{-u}) \quad [\text{logistic CDF}]$$

$$\ln[Pr(D_i^* = 1) / (1 - Pr(D_i^* = 1))] = \beta_0 + \beta_1 \theta_i + \beta_2 s_i + \beta_3 \sigma_i + \beta_4 \mu_i \quad [\text{log-odds form}]$$

$$OR(\theta_i) = \exp(\beta_1) \quad [\text{odds ratio; } OR > 1 \text{ implies } \theta_i \text{ increases export odds}]$$

where:

- D_i^* = binary export participation indicator: 1 if total exports_i > 0, else 0
- $\Lambda(\cdot)$ = logistic cumulative distribution function
- θ_i = royalty intensity [technology acquisition proxy; expected sign: positive]
- s_i = log total assets
- σ_i = software intensity
- μ_i = imported raw material intensity
- $OR(\theta_i) = \exp(\beta_1)$: odds ratio for royalty intensity

Logistic regression is appropriate because D_i^ is binary. Maximum likelihood estimation ensures predicted probabilities are bounded within (0, 1). A positive β_1 ($OR > 1$) would indicate that technology-licensing firms face higher odds of export market participation.*

Logistic regression is used because the dependent variable — whether the firm exported — is binary. A positive coefficient (odds ratio greater than one) on royalty intensity would indicate that firms paying for licensed foreign technology are more likely to export, consistent with technology licensing improving product quality and providing access to the licensor's distribution networks and quality standards.

4.3.4 Service Export Probability Model (Logit)

Equation B.4 — Service Export Probability Model (Logit)

$$Pr(D_i^{se} = 1) = \Lambda(\beta_0 + \beta_1 \rho_i + \beta_2 \sigma_i + \beta_3 IT_i + \beta_4 s_i) \quad (B.4)$$

where:

- D_i^{se} = net service exporter dummy: 1 if services exports_i > goods exports_i, else 0
- ρ_i = R&D intensity [Formula A.6; expected sign: positive]
- σ_i = software intensity [Formula A.8]
- IT_i = IT-enabled services intensity: ITES charges_i / (total income_i + 1)
- s_i = log total assets [firm-size control]

A positive and significant β_1 would indicate that innovation capability (R&D) is more closely associated with knowledge-intensive services trade than with goods trade, consistent with the hypothesis that service exports depend on accumulated organisational capabilities.

Logistic regression is used because the dependent variable — whether a firm's service exports exceed its goods exports — is binary. R&D intensity is expected to carry a positive

coefficient on the premise that service exports, such as information technology, consulting, research, and financial services, are knowledge-intensive and depend on the capabilities that R&D investment builds. This is of particular relevance to India, where IT and business services represent a substantial and growing share of total exports.

4.3.5 R&D × Technology Sector Interaction Model (OLS)

Equation B.5 — R&D × Technology-Sector Interaction Model (OLS)

$$\tau_i = \beta_0 + \beta_1 \rho_i + \beta_2 T_i + \beta_3 (\rho_i \times T_i) + \beta_4 s_i + \beta_5 \sigma_i + \beta_6 \mu_i + \varepsilon_i \quad (\text{B.5})$$

$$\text{Marginal effect of R\&D: } \partial \tau_i / \partial \rho_i = \beta_1 + \beta_3 T_i$$

where:

τ_i = export intensity [dependent variable]

ρ_i = R&D intensity

T_i = technology-sector dummy: 1 if NIC $\in \{26, 62, 72\}$, else 0

$\rho_i \times T_i$ = interaction term: product of R&D intensity and technology-sector dummy

β_1 = marginal R&D–export association for non-technology-sector firms ($T_i = 0$)

β_2 = export intensity level difference between technology and other sectors, holding $\rho_i = 0$

β_3 = differential R&D effect in technology sectors (key parameter; positive if R&D more effective in tech sectors)

$\beta_4 - \beta_6$ = coefficients on size, software intensity, and imported raw material intensity

When $T_i = 0$ (non-technology sector): $\partial \tau_i / \partial \rho_i = \beta_1$. When $T_i = 1$ (technology sector): $\partial \tau_i / \partial \rho_i = \beta_1 + \beta_3$. A significant positive β_3 confirms that R&D has a stronger export-intensity effect in electronics, software/IT, and R&D services relative to other industries.

OLS is used as the dependent variable is continuous. A positive and significant interaction coefficient (β_3) would indicate that, for a given level of R&D spending, technology-sector firms gain more in export intensity than firms in other sectors.

4.3.6 Digital Intensity Models (OLS and Logit)

Equation C.1 — Export Intensity on Digital Intensity Index (OLS)

$$\tau_i = \beta_0 + \beta_1 DII_i + \beta_2 s_i + \beta_3 RD_i + \beta_4 \mu_i + \beta_5 W_i + \varepsilon_i \quad (\text{C.1})$$

where:

τ_i = export intensity [dependent variable]

DII_i = Digital Intensity Index for firm i [see Equation A.11; expected sign: positive]

s_i = log total assets [firm-size control]

RD_i = R&D expenditure of firm i, in INR million [entered in levels]

μ_i = imported raw material intensity

W_i = salary and wage expenditure of firm i, in INR million [human capital proxy]

β_1 = association between DII and export intensity; 2020–2025 estimate: -0.011 ($p = 0.954$, n.s.)

Equation C.2 — Export Participation on Digital Intensity Index (Logit)

$$Pr(D_i^* = 1) = A(\beta_0 + \beta_1 DII_i + \beta_2 TA_i + \beta_3 RD_i + \beta_4 IRM_i + \beta_5 W_i) \quad (C.2)$$

$$\ln[\pi_i / (1 - \pi_i)] = \beta_0 + \beta_1 DII_i + \beta_2 TA_i + \beta_3 RD_i + \beta_4 IRM_i + \beta_5 W_i \quad [\text{log-odds form}]$$

where:

$\pi_i = \Pr(D_i^* = 1)$: predicted probability of export participation for firm i

DII_i = Digital Intensity Index; estimated coefficient: +0.141 ($p = 0.541$, n.s.)

TA_i = total assets of firm i in levels (INR million)

RD_i = R&D expenditure; estimated coefficient: -0.097 ($p = 0.457$, n.s.)

IRM_i = imported raw materials in levels; estimated coefficient: +0.057 ($p < 0.001$)***

W_i = salary and wage expenditure in levels; estimated coefficient: +0.015 ($p = 0.009$)**

β_0 = intercept; estimated: -1.442 ($p < 0.001$)***

*Pseudo- R^2 (McFadden) = 0.138. Likelihood ratio test: $p < 0.001$. $N = 473$ (2014–2019 sample). Significance: *** $p < 0.001$, ** $p < 0.01$.*

OLS is used for the continuous export intensity outcome and logistic regression for the binary export participation outcome. The Digital Intensity Index is expected to carry a positive coefficient under the hypothesis that digital infrastructure investment reduces trade costs and facilitates cross-border transactions, while imported raw material intensity and salary expenditure are expected to be positively associated with export participation, reflecting global value chain integration and workforce skill respectively.

4.3.7 Digital × Services Interaction Model (OLS)**Equation C.4 — Digital Intensity × Services Interaction Model (OLS)**

$$\tau_i = \beta_0 + \beta_1 DII_i + \beta_2 S_i + \beta_3 (DII_i \times S_i) + Controls_i + \varepsilon_i \quad (C.4)$$

$$\text{Marginal effect of DII: } \partial\tau_i / \partial DII_i = \beta_1 + \beta_3 S_i$$

where:

τ_i = export intensity [dependent variable]

DII_i = Digital Intensity Index [Equation A.11]

S_i = services dummy: 1 if NIC ≥ 5000 , else 0 (manufacturing)

$DII_i \times S_i$ = interaction term

β_1 = digital–export association for manufacturing firms ($S_i = 0$)

β_2 = export intensity level difference between services and manufacturing, holding $DII_i = 0$

β_3 = differential digital effect in services relative to manufacturing [2014–19: -0.350, $p = 0.435$ n.s.; 2020–25: +0.001, $p = 0.996$ n.s.]

When $S_i = 0$ (manufacturing): $\partial\tau_i / \partial DII_i = \beta_1$. When $S_i = 1$ (services): $\partial\tau_i / \partial DII_i = \beta_1 + \beta_3$. The non-significant interaction coefficient in both periods provides no support for the hypothesis that services firms derive greater export returns from digital investment than manufacturing firms.

OLS is used as the dependent variable is continuous. A positive and significant interaction coefficient would indicate that digital investment is more strongly associated with export

intensity in services firms than in manufacturing firms, consistent with digital tools being more directly embedded in the production and delivery of services.

4.3.8 Treatment of Unobserved Heterogeneity

Industry fixed effects are included using two-digit NIC codes throughout. The underlying dataset does not consistently provide firm-level identifiers suitable for the construction of a balanced panel, and firm fixed effects are therefore not employed in any of the models presented in this study. As a result, time-invariant firm-level characteristics — such as management quality, brand reputation, or historical relationships with foreign buyers — are not directly controlled for. Industry and period fixed effects partially address this concern by absorbing variation common to firms within the same industry and time period, but they cannot account for heterogeneity that varies across firms within an industry-period cell. The implications of this limitation for causal interpretation are discussed further in Section 7.1.

SECTION 5: ANALYSIS & FINDINGS

This section reports the empirical results, organised by research question, followed by an integrated discussion and an assessment of economic significance. Coefficient-level estimates and standard errors are reported as obtained from the underlying regression output; where estimates were not retained in the underlying analysis, this is stated transparently rather than reconstructed.

5.1 Research Question 1: R&D, Technology, and Export Performance

5.1.1 Log Exports Model

Table 2 reports the model fit statistics for the regression of log exports on log R&D spending, log total assets, software intensity, imported raw material intensity, and industry dummies. The coefficient on log R&D is positive and statistically significant in the 2020–2025 sample (0.0642, SE = 0.021, $t = 3.044$, $p = 0.002$) but not in the smaller 2014–2019 sample (0.1579, SE = 0.231, $t = 0.684$, $p = 0.494$).

Table 2. OLS Model Summary — Log Exports on R&D and Controls

Model Statistic	2020–2025 (N = 8,808)	2014–2019 (N = 473)
R ²	0.113	0.313
Adjusted R ²	0.106	0.253
F-statistic	15.73***	5.21***
Coefficient — ln(R&D)	0.0642**	0.1579 (n.s.)
Standard Error — ln(R&D)	0.021 (t = 3.044, p = 0.002)	0.231 (t = 0.684, p = 0.494)
Industry Fixed Effects	Yes	Yes
AIC / BIC	3.176e+04 / 3.227e+04	1,686 / 1,848
Intercept	−0.14 (p = 0.289)	−1.89 (p = 0.060)

Note: *** $p < 0.001$, ** $p < 0.01$. Model fit statistics are taken directly from the underlying regression output.

Table 3 reports selected industry coefficients from the 2020–2025 Log Exports Model. Pharmaceuticals, apparel, food and beverages, rubber and plastics, and chemicals all carry positive and statistically significant coefficients, while construction-related industries carry negative and statistically significant coefficients.

Table 3. Selected Industry Coefficients — Log Exports Model (2020–2025)

Industry (NIC Code)	Coefficient	Significance
Food and beverages (NIC 15)	1.06	p = 0.010
Apparel (NIC 14)	0.60	p < 0.001
Pharmaceuticals (NIC 21)	0.66	p < 0.001
Rubber and plastics (NIC 22)	0.43	p < 0.01
Chemicals (NIC 20)	0.26	p = 0.049
Construction (NIC 41)	−0.42	p = 0.019
Construction (NIC 42)	−0.53	p < 0.01

Note: the omitted reference category is the lowest-numbered industry group in the sample (NIC 10); standard errors for individual industry coefficients were not separately reported in the underlying analysis.

5.1.2 Export Participation Model: Royalty Intensity (Logit)

Contrary to the hypothesised positive relationship, the coefficient on royalty intensity is negative in both periods and is not statistically significant in either: −0.0415 (SE = 1.412, $z = -0.029$, $p = 0.977$; odds ratio = 0.959) in the 2020–2025 sample and −22.8454 (SE = 53.792, $z = -0.425$, $p = 0.671$; odds ratio ≈ 0) in the 2014–2019 sample. No reliable conclusion can be drawn about the direction of the relationship between royalty payments and export participation from this analysis.

Table 4. Logit Model — Export Participation on Royalty Intensity

Variable	2020–2025 (N = 8,808)	2014–2019 (N = 473)
Royalty Intensity	−0.0415 (SE 1.412, p = 0.977)	−22.8454 (SE 53.792, p = 0.671)

5.1.3 Service Export Probability Model

In the 2020–2025 sample (N = 8,808), R&D intensity carries a negative, non-significant coefficient of −1.0567 (p = 0.798), while software intensity (23.4943, p < 0.001) and IT services intensity (17.5892, p = 0.003) are both positive and statistically significant, alongside firm size (0.0897, p = 0.009). In the 2014–2019 sample (N = 473), none of the explanatory variables are statistically significant except firm size (0.8794, p = 0.001); R&D intensity is −10.4802 (p = 0.893), software intensity is −0.5354 (p = 0.999), and IT services intensity is 111.8937 (p = 0.226).

Table 5. Logit Model — Service Export Probability (2020–2025)

Variable	Coefficient	p-value	Significance
R&D Intensity	−1.0567	0.798	Not Significant
Software Intensity	23.4943	< 0.001	*** Significant
IT Services Intensity	17.5892	0.003	** Significant
Firm Size (ln Assets)	0.0897	0.009	** Significant

5.1.4 R&D × Technology Sector Interaction Model

This model tests whether the R&D–export intensity relationship differs between technology-intensive sectors and other industries. In the 2020–2025 sample, none of the relevant coefficients are statistically significant: R&D intensity (0.0002, p = 0.999), the technology-sector dummy (0.0247, p = 0.677), and the interaction term (−0.0247, p = 0.954). In the 2014–2019 sample, the coefficients are larger in magnitude but remain non-significant: R&D intensity (1.9141, p = 0.226), the technology-sector dummy (0.0202, p = 0.510), and the interaction term (−2.9510, p = 0.131). The data therefore do not support the hypothesis that R&D investment is more productive for export intensity in technology-intensive sectors.

Table 6. OLS Model — R&D × Technology Sector Interaction (2020–2025)

Variable	Coefficient	p-value	Significance
R&D Intensity	0.0002	0.999	Not Significant
Technology Sector Dummy	0.0247	0.677	Not Significant
R&D Intensity × Technology Sector	−0.0247	0.954	Not Significant

5.2 Research Question 2: Digital Infrastructure and Export Competitiveness

5.2.1 Digital Intensity Index and Export Intensity (OLS)

Table 7. OLS Model — Export Intensity on Digital Intensity Index

Variable	2020–2025 Coefficient	2014–2019 Coefficient
Digital Intensity Index	–0.011 (p = 0.954, n.s.)	0.0190 (SE 0.021, t = 0.905, p = 0.366, n.s.)

5.2.2 Software Spending and Service Exports (OLS)

An OLS model uses service exports as the dependent variable and software charges as the key predictor. The coefficient on software charges is not statistically significant in either period: 0.0307 (p = 0.659) in the 2020–2025 sample (N = 8,806) and –9.8990 (p = 0.615) in the 2014–2019 sample (N = 471). In both samples, salary expenditure is a significant positive predictor of service exports, and in the 2014–2019 sample total assets is also significant (0.0125, p < 0.001).

Table 8. OLS Model — Service Exports on Software Spending

Variable	2020–2025 Coefficient	2014–2019 Coefficient
Software Charges	0.0307 (p = 0.659, n.s.)	–9.8990 (SE 19.656, p = 0.615, n.s.)

5.2.3 Digital Intensity and Export Participation (Logit, 2014–2019)

Table 9. Logit Model — Export Participation on Digital Intensity, 2014–2019

Variable	Coefficient	p-value	Significance
Intercept	–1.442	< 0.001	*** Significant
Digital Intensity Index	0.141	0.541	Not Significant
Total Assets	2.31e ^{–5}	0.746	Not Significant
R&D Expenses	–0.097	0.457	Not Significant
Imported Raw Materials	0.057	< 0.001	*** Significant
Salaries and Wages	0.015	0.009	** Significant

Note: Pseudo R² = 0.138. Likelihood ratio test p < 0.001. N = 473 (2014–2019 sample).

5.2.4 Digital × Services Interaction Model

Table 10. OLS Model — Digital Intensity × Services Interaction

Model Statistic	2014–2019	2020–2025
F-statistic	3.786***	0.0327
p-value (overall model)	0.0005	1.00
R ²	0.054	0.000
Interaction coefficient (Digital Index × Services)	–0.350	0.001
p-value (interaction)	0.435 (n.s.)	0.996 (n.s.)

5.3 Discussion of Findings

This section interprets the results reported in Sections 5.1 and 5.2 with reference to the theory and literature reviewed in Section 2, focusing primarily on variables that are statistically significant.

5.3.1 Digital Intensity

The coefficient on the Digital Intensity Index is not statistically significant in either the export intensity model (2020–2025 coefficient = –0.011, $p = 0.954$) or the export participation model (2014–2019 coefficient = 0.141, $p = 0.541$). The evidence indicates that, on average, firms with higher digital spending do not exhibit significantly higher export intensity or export probability. One plausible explanation is a basic-capability argument: digital infrastructure may have become a standard feature of most actively trading firms, particularly after 2020, such that differences in digital spending no longer meaningfully predict differences in export performance. This pattern is consistent with reduced cross-firm variation in digital adoption following the COVID-19 pandemic, which would compress the explanatory power of digital spending as a continuous predictor. A further consideration concerns measurement: the Digital Intensity Index is constructed from expenditure data, which captures inputs into digital infrastructure rather than the firm’s actual digital capability or the effectiveness with which digital tools are deployed, and this distinction may attenuate the estimated relationship between digital intensity and export outcomes.

5.3.2 R&D Intensity

The results suggest a directionally positive but generally statistically weak relationship between R&D intensity and export outcomes across most of the models in which R&D-related coefficients are reported (for example, the non-significant R&D Expenses coefficient of –0.097, $p = 0.457$, in the digital intensity logit model), with the notable exception of the Log Exports Model, where the 2020–2025 coefficient on log R&D is positive and statistically significant. Several explanations are consistent with this overall

pattern, several of which echo the mixed findings reported in Section 2.2. First, R&D investment is typically characterised by long gestation periods between expenditure and commercialisation, such that the export effects of R&D spending in a given year may not be fully realised within the same period — a delayed-commercialisation effect that a cross-sectional or short-panel specification may not fully capture, consistent with the pre-export investment timing documented by Aw, Roberts, and Winston (2007). Second, R&D expenditure among Indian firms is highly concentrated in a small number of sectors — pharmaceuticals, chemicals, and information technology — which limits the generalisability of an average R&D coefficient estimated across all industries, consistent with the sector-contingent findings of Montobbio and Rampa (2005). Third, R&D may operate on exports indirectly, for instance through product quality improvements that raise overall firm competitiveness rather than acting as a direct, immediate determinant of export volume, consistent with Basant, Mishra, and Saha's (2016) finding of no significant exporter/non-exporter difference in technology spending among firms without foreign affiliates.

5.3.3 Imported Raw Material Intensity

Imported raw material intensity is positive and statistically significant in the export participation logit model (coefficient = 0.057, $p < 0.001$). This finding is consistent with the global value chains literature, which suggests that firms already sourcing inputs internationally are better integrated into international supplier networks and face lower informational and relational barriers to entering export markets. The evidence is also consistent with a learning-by-importing mechanism, whereby exposure to foreign suppliers transmits knowledge of international quality standards and production practices that subsequently support export readiness. From a production-fragmentation perspective, firms integrated into import-based supply chains may already operate at the quality and compliance standards required by foreign buyers, reducing the marginal cost of beginning to export.

5.3.4 Salary and Wage Expenditure

Salary and wage expenditure is positive and statistically significant in the export participation logit model (coefficient = 0.015, $p = 0.009$) and is also a significant positive predictor of service exports in both sample periods (Section 5.2.2). This result is consistent with a human capital interpretation, in which higher wage expenditure proxies for a more skilled workforce, which is plausibly associated with greater absorptive capacity — the ability of a firm to identify, assimilate, and apply external knowledge, including knowledge embedded in export market requirements. The finding is also consistent with a technology-complementarity argument, under which digital and technological investments are more effective for exporters when paired with a workforce

capable of using them productively, which may help explain why salary expenditure remains significant even as the Digital Intensity Index itself does not.

5.4 Economic Significance

This section discusses the economic magnitude of the statistically significant coefficients reported above, in addition to their statistical significance, consistent with the principle that statistical significance alone does not establish managerial or policy relevance.

Imported Raw Material Intensity: in the export participation logit model, the coefficient of 0.057 implies that a one-unit increase in imported raw material intensity is associated with an increase in the odds of exporting by a factor of approximately $e^{0.057} \approx 1.059$, or roughly 5.9 percent. While the precise scale of imported raw material intensity is not specified in the underlying analysis, the direction and statistical reliability of the effect indicate that supply chain integration is among the more economically meaningful predictors of export participation identified in this study.

Equations D.1 & D.2 — Odds Ratio and Percentage-Change Interpretation

$$OR_k = \exp(\beta_k) \quad [\text{odds ratio for predictor } k] \quad (D.1)$$

$$\Delta \text{ Odds } (\%) = 100 \times [\exp(\beta_k) - 1] \quad [\text{percentage change in export odds}] \quad (D.1a)$$

$$\% \Delta \approx 100 \times [\exp(\beta_k) - 1] \quad [\text{log-point coefficient to percentage, OLS}] \quad (D.2)$$

where:

OR_k = odds ratio for predictor k: multiplicative change in export odds per unit increase in x_k

β_k = logit or OLS coefficient estimated by maximum likelihood or least squares

$\exp(\cdot)$ = exponential function (base e)

Empirical results: Imported raw materials: $\exp(0.057) \approx 1.059$, implying a 5.9% increase in export odds per unit increase in μ_i . Salary expenditure: $\exp(0.015) \approx 1.015$, implying a 1.5% increase in export odds per unit increase in W_i . Industry effect (food & beverages, NIC 15): $100 \times [\exp(1.06) - 1] \approx 188\%$ higher exports than the reference sector.

Salaries and Wages: the coefficient of 0.015 implies an increase in the odds of exporting by a factor of approximately $e^{0.015} \approx 1.015$, or roughly 1.5 percent, for a one-unit increase in salary expenditure as measured in the model. Although smaller in magnitude than the imported raw materials effect, this result is statistically reliable and points to workforce quality as a consistently relevant, if modest, correlate of export participation.

Digital Intensity Index and R&D Expenses: because the coefficients on the Digital Intensity Index and R&D Expenses are not statistically significant in most of the models in which they are reported, their economic significance cannot be reliably established from the current analysis. The corresponding point estimates (−0.011 and −0.097, respectively,

in the relevant models) are reported for completeness, but given their associated p-values (0.954 and 0.457), no economically meaningful interpretation is warranted from these specific coefficients.

Equations D.3 & D.4 — Logit Model Fit Statistics

$$R_m^2 = 1 - [\ell(\hat{\beta}) / \ell(0)] \quad [\text{McFadden pseudo-}R^2] \quad (\text{D.3})$$

$$LR = -2 [\ell(0) - \ell(\hat{\beta})] \sim \chi^2(df) \quad [\text{Likelihood Ratio test}] \quad (\text{D.4})$$

where:

R_m^2 = McFadden pseudo- R^2 : proportionate reduction in log-likelihood from null to estimated model

$\ell(\hat{\beta})$ = log-likelihood of the estimated logit model

$\ell(0)$ = log-likelihood of the null (intercept-only) model

LR = likelihood ratio test statistic, asymptotically distributed χ^2 with df equal to number of predictors

Reported values: pseudo- $R^2 = 0.138$ for Model C.2 (export participation on digital intensity, 2014–2019).

Likelihood ratio test: $p < 0.001$.

Industry Effects: the industry coefficients in the Log Exports Model are economically large relative to the imported raw materials and salary effects — for example, the food and beverages coefficient of 1.06 implies exports approximately $e^{1.06} - 1 \approx 188$ percent higher than the reference industry, holding other factors constant. This suggests that sectoral classification carries substantially greater economic weight in explaining cross-firm differences in export volume than the firm-level technology variables of primary interest in this study.

SECTION 6: IMPLICATIONS OF RESEARCH

6.1 Implications for Corporate Strategy and Policy

6.1.1 Implications for Firms and Export Managers

- Firms in the pharmaceuticals, chemicals, apparel, and food and beverages sectors exhibit a structural export advantage. Managers in these sectors may benefit from continued investment in product quality and R&D to reinforce their position in global markets. For firms in sectors with low or negative export coefficients, such as construction, R&D and digital investment alone are unlikely to generate export opportunities without first establishing a globally competitive product or service offering.
- The statistically significant relationship between imported raw materials and export participation suggests that firms considering entry into export markets may benefit from examining their supply chain. Sourcing inputs from global suppliers can

improve product standards, build international contacts, and expose firms to the requirements of foreign buyers, potentially offering a more cost-effective pathway to export readiness than large upfront R&D investments.

- For firms that pay royalties or license foreign technology, the evidence remains directionally consistent with improved export capability even though the relevant coefficients are not statistically significant in this sample. Technology licensing may be understood not only as a production input but also as a market-access strategy, given its theoretical association with product quality improvement and access to a licensor's distribution networks.
- From a corporate finance perspective, the decision to upgrade equipment or information-technology infrastructure when entering international markets should be weighed against the firm's ability to manage currency risk and to demonstrate a return on investment within the new market environment; competitive pressure from international rivals makes some degree of technology upgrading a prudent strategic default, even where the average statistical association between aggregate technology spending and export outcomes is weak.

6.1.2 Implications for Digital Investment Decisions

- The absence of a statistically significant relationship between digital spending and export performance indicates that high digital expenditure alone does not guarantee improved export outcomes; it may instead reflect a baseline level of digitalisation that is now common among trading firms. Managers may therefore consider focusing less on the overall magnitude of digital spending and more on the specific allocation of digital investment — for example, e-commerce platforms, logistics tracking, or customs compliance software — which are more directly tied to trade cost reduction.
- Because salary expenditure is consistently significant while the Digital Intensity Index is not, workforce quality appears to be a relevant complement to digital investment. Digital tools are plausibly most effective when paired with employees who possess the skills to use them, suggesting that digital infrastructure upgrades may be more effective when accompanied by investment in workforce training.

6.1.3 Implications for Policymakers

- The sectoral variation in export coefficients suggests that uniform, sector-blind innovation subsidies may not be the most efficient policy instrument. Support directed at sectors with an established export structure — pharmaceuticals, chemicals, and IT services — may generate larger returns than broad-based incentives applied uniformly across industries.

- The statistically significant association between imported raw material intensity and export participation points to the potential value of trade facilitation policies, such as reducing tariffs on intermediate goods used in export production and simplifying import-export procedures, as a complement to supply-side innovation subsidies.
- Foreign buyers, particularly in business-to-business markets, typically impose more demanding product-quality, certification (e.g., ISO), and technology requirements than domestic markets; export-promotion policy may therefore usefully combine innovation support with technical assistance for quality certification and standards compliance, rather than R&D subsidy alone.

6.2 Methodological Implications

The findings of this study carry several methodological implications for future firm-level research on technology and exports in India.

- Selection bias remains a central concern. Because exporters are systematically larger, more productive, and more capital-intensive than non-exporters, future studies should, where feasible, employ propensity score matching or fixed-effects panel methods that compare similar firms before and after export entry, rather than relying solely on pooled cross-sectional comparisons of the kind used in this study.
- Instrumental variable (IV) approaches would help address the endogeneity concern discussed in Section 7.1, in which exporting firms may invest more in R&D and digital infrastructure as a consequence of competitive exposure rather than the reverse; without such an approach, the coefficients reported in Section 5 should continue to be interpreted as associations rather than causal effects.
- Where consistent firm identifiers become available, firm fixed-effects panel models would allow researchers to control for time-invariant firm-level heterogeneity — such as management quality or historical buyer relationships — strengthening the internal validity of estimated relationships beyond what industry and period fixed effects can achieve.
- Given the long gestation periods associated with R&D investment documented in Section 2.2, dynamic panel models incorporating lagged R&D and digital intensity variables could better capture delayed effects on export performance than the contemporaneous specifications used in this study.
- Incorporating data on export destination and trading-partner characteristics would allow future research to examine whether the relationships identified in this study vary across different export markets, complementing the aggregate measures of export intensity and participation used here.

SECTION 7: CONCLUSION

7.1 Limitations and Future Research

7.1.1 Data Limitations

CMIE Prowess primarily covers large and medium-sized listed companies. Small and medium enterprises (SMEs), which constitute a substantial share of India's export base in labour-intensive sectors, are largely absent from the database. The results therefore reflect the behaviour of larger, more established firms and may not generalise to the broader population of exporting firms. The 2014–2019 sample (473 observations) is substantially smaller than the 2020–2025 sample (8,808 observations); this asymmetry limits the comparability of results across periods and renders some sector-level estimates in the earlier period less reliable due to small cell sizes. The data also do not include export destination, product category, or trading-partner information, which restricts the analysis to aggregate measures of export intensity and participation.

7.1.2 Measurement Error

The Digital Intensity Index is constructed from spending data, which captures inputs rather than realised digital capability. A firm may spend heavily on information technology while deploying it inefficiently, while another firm may achieve comparable or superior digital performance at lower expenditure. R&D expenditure similarly does not always translate into usable innovation outputs. These measurement gaps plausibly attenuate the estimated coefficients on R&D and digital intensity. Alternative measures, such as patent filings, product innovation counts, or digital adoption scores, would improve the precision of the analysis but are not available within CMIE Prowess.

7.1.3 Absence of Firm Fixed Effects and Endogeneity

The models in this study are estimated using pooled OLS and logit specifications, without firm fixed effects, for the reasons discussed in Section 4.3.8. There is also a plausible endogeneity concern: firms that already export may invest more in R&D and digital infrastructure as a consequence of their exposure to international competition, rather than R&D and digital investment causing the firm to begin exporting. Without an instrument or a natural experiment, it is not possible to establish a causal direction from the current analysis, and the reported coefficients should be interpreted as associations rather than causal effects.

7.1.4 Recommendations for Future Research

Building on Section 6.2, future research employing instrumental variables, firm fixed-effects panel models, dynamic panel specifications, and export-destination data would substantially strengthen the causal interpretation of the relationships identified in this study.

7.2 Concluding Remarks

Read as a whole, the literature reviewed in Section 2 traces a narrative, spanning two decades of research from Wagner (2007) to Sahoo, Vadakoot Sathees, and Anand (2025), in which technological capability is positioned as a central determinant of Indian firms' ability to compete in global markets. The empirical findings of this study offer an important qualification to that narrative. Among the large sample of Indian firms examined here, R&D intensity and the Digital Intensity Index exhibit directionally positive but generally statistically weak associations with export outcomes, while sectoral classification, imported raw material intensity, and salary expenditure emerge as the most consistent and economically meaningful predictors of export performance.

This does not refute the broader literature's conclusion that technology matters for export competitiveness; rather, it suggests that the relationship is highly context-dependent — varying with the industry in which a firm operates, the extent to which it is already integrated into global supply chains through imported inputs, and the skill level of its workforce — rather than operating as a uniform, stand-alone driver of export success. These findings point to a need for more targeted innovation and export policies that account for the structural differences across Indian industries, rather than uniform incentives applied without regard to sectoral or firm-level heterogeneity. They also point, more specifically, to the continued importance of global value chain integration and human capital investment as levers that firms and policymakers can use to support export competitiveness, alongside — rather than in place of — conventional R&D and digital infrastructure investment.

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